Direct air capture of CO2 at a climate-relevant scale

An evidence review for a regional committee deciding on pilot funding

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Summary

Direct air capture (DAC) works as chemistry, but it does not yet work at a scale that matters for the climate. All verified DAC plants together removed roughly 1,500 tonnes of CO2 in 2025, about six orders of magnitude below the ~1 gigatonne per year (Gt/yr) benchmark that net-zero pathway assessments imply for mid-century. Flagship plants run far below design: the largest operating plant, Climeworks' Mammoth (36,000 t/yr design capacity), removed 675 tonnes in the first half of 2026 on the company's own figures, and the largest plant yet built, Occidental's 500,000 t/yr Stratos in Texas, had still not begun commercial operations in late 2026 after roughly two years of delay. Costs actually paid today run from several hundred dollars up to ~$1,300 per tonne, against independent projections of $230-540/t by 2050; the widely cited $100/t goal looks implausible on present evidence. There is no fundamental thermodynamic or storage-capacity barrier at the benchmark scale, but the implied build rate (tens of 1-Mt plants per year), the additional clean energy (~5-10% of current world electricity per Gt/yr), and a fragile, highly concentrated demand base make the pathway extremely demanding. For a funding committee, the evidence supports only conditional, verification-first pilot funding: pay for delivered, independently verified tonnes with durable storage and additional clean energy, treat pilots as learning investments, and expect far more verified removals per euro from cheaper conventional options in the near term.

Scope, concepts and questions

Direct air capture uses engineered systems to extract CO2 from ambient air; when the captured CO2 is stored durably (usually underground), the result is durable carbon dioxide removal (CDR). This review answers one primary and five secondary questions posed by the requesting brief:

- Q0: Can DAC operate at a scale that materially affects atmospheric CO2 and climate goals?
- Q1: How does DAC work in engineering terms, and what does it require?
- Q2: What is the current and announced deployment scale, against the climate-relevant benchmark?
- Q3: What do current and projected costs per tonne look like, and how credible are cost-reduction pathways?
- Q4: What dependencies and integrity risks constrain deployment at meaningful scale?
- Q5: What does the evidence imply for a committee deciding whether to fund DAC pilots, compared with alternative uses of the same funds?

"A scale that matters" is taken from IPCC/IEA-style net-zero pathway assessments. In the IEA Net Zero Emissions by 2050 Scenario, DAC captures more than 85 Mt CO2 in 2030 and around 980 Mt in 2050, growing from ~0.01 Mt today [1] , consistent with the ~1 Gt/yr mid-century benchmark used here. Total CDR needed is far larger: in the highest-ambition scenario of the 2026 State of CDR assessment, global CDR must grow from 2.2 Gt/yr today to ~8.8 Gt/yr by 2050, and "novel" CDR (biochar, DAC, and others beyond forestry and soils) must reach gigatonne scale by 2050 in every scenario examined [6] . For context, fossil fuels emitted a record ~38.1 Gt CO2 in 2025 [5] , and, in climate terms, removing and durably storing one tonne is equivalent to not emitting it [3] .

Review method, evidence base and stopping

This is a structured narrative review of publicly accessible sources inspected up to 9 November 2026: intergovernmental assessments (IEA), national-academy analyses (NASEM), peer-reviewed studies (Joule, Nature Communications), official company and agency statements (Climeworks, Occidental subsidiaries, DOE actions via specialized press), and reputable independent syntheses (Carbon Brief on the State of CDR report; CDR-market compilations drawing on registry data). Demonstrated, nameplate and projected performance are kept distinct throughout. Several primary documents were inaccessible (account-walled IEA report PDF, blocked journal sites, front matter only of the State of CDR PDF); in those cases inspected secondary accounts are cited as such. Searches stopped when repeated query families returned only known or low-relevance material. This report's PDF was verified structurally and textually; visual page inspection was not available (a declared capability gap of the producing environment).

Q1. How DAC works and what it requires

The intuitive problem is dilution: CO2 is only ~0.04% of air, roughly 300 times more dilute than flue gas from a coal plant, so a machine must move enormous volumes of air to harvest each tonne, and thermodynamics sets an irreducible energy floor for separating a dilute mixture [3] . All engineered designs share three steps: fans push air over a capture material; the material is regenerated by releasing the CO2 (usually with heat); the released CO2 is compressed, transported and stored.

Two designs dominate. Liquid-solvent systems (the Carbon Engineering design, now owned by Occidental) pass air through a potassium hydroxide solution and regenerate it in a high-temperature calciner; the reference engineering design requires either 8.81 GJ of natural gas, or 5.25 GJ of gas plus 366 kWh of electricity, per tonne captured [2] . Solid-sorbent systems (Climeworks) push air through amine-coated filters regenerated at low temperature, "like boiling water" in the company's geothermal setting [13] . Electrochemical routes are at an earlier stage [3] .

F1. Total energy demand, combining heat and electricity, is the dominant physical requirement: current systems use on the order of 1,500-3,000 kWh per tonne captured and stored [5] . Energy is also the dominant integrity risk: the net reduction delivered depends on the energy source assumed (renewables, nuclear, gas or coal) [3] , so gas-fired designs must recapture their own combustion CO2, as the reference design does [2] . Water and land footprints are comparatively modest: the IEA describes DAC's land and water footprint as "relatively small" among CDR approaches, and impact assessments find the on-site resource use of a large DAC plant comparable to other industrial infrastructure, with chemical production of solvents and sorbents a project-specific consideration [1] [16] . Amine sorbents and caustic solutions are consumed and replaced over time, so sorbent manufacture enters the life-cycle account [16] .

Q2. Deployment scale against the benchmark

F2. DAC today operates roughly six orders of magnitude below the climate-relevant benchmark. Eighteen small plants were operating in 2022 [1] ; the combined design capacity of all operating facilities was under 50,000 t/yr in April 2026 [10] ; and verified removals (the tonnes actually delivered and recorded, rather than nameplate) totalled about 1,500 t in 2025, against 38.1 Gt of fossil emissions [5] . That is about one tonne removed for every 25 million tonnes emitted.

F3. Nameplate and delivery diverge sharply, which is why the two must not be conflated. Orca (4,000 t/yr rating) captured at most ~1,000 t in any year. Mammoth (36,000 t/yr design) removed 675 t in the first half of 2026 on company figures, under 4% of its design rate, with only 12 of 72 collectors running in mid-2025 while faults were fixed; the company says early-2026 output was more than five times a year earlier and unit operating cost more than halved, but these are unaudited company figures [5] [13] . The largest plant yet built, Occidental's 500,000 t/yr Stratos, under construction since May 2023, was slated to start in late 2024 and was still in startup in 2026, with the operator guiding to

commercial operation around end-2026 [7] [8] .

The announced pipeline is likewise modest against the benchmark and has been shrinking and slipping. The US DOE's $3.5bn DAC-hubs programme cancelled 10 of 21 projects in October 2025 (part of ~$7.5bn in terminated awards), before retaining $1.2bn in April 2026 for the two flagship hubs, Project Cypress (1 Mt/yr design, with Climeworks and Heirloom, already delayed alongside Climeworks' 2025 cost cuts) and the South Texas hub [9] [10] [12] . Corporate ambitions illustrate the delivery gap: Climeworks once targeted 225 Mt/yr by 2025 (missed by ~150,000x) and still targets megaton capacity by 2030 and gigaton scale by 2050 [5] [13] . Against the IEA pathway (>85 Mt by 2030), the gap is a factor of ~1,700 from today's global nameplate, reaching ~980 Mt by 2050 would require building capacity equivalent to ~40 new 1-Mt plants every year from now to 2050 (this is my arithmetic from the IEA figures, not an IEA statement) [1] . Novel CDR as a whole was ~2 Mt/yr and growing ~40% per year, a solar-like growth rate, yet still nowhere near the gigatonne scale needed by 2050 [6] . DAC is 0.07% of novel CDR [5] .

Table 1. The scale ladder: today vs the benchmark (values from sources as cited).

Scale item	Status	Volume (t CO2/yr)
Verified global DAC removal, 2025	Delivered	~1,500 (per year)
All operating plants combined, nameplate (Apr 2026)	Operating	< 50,000
Largest operating plant (Mammoth), design / actual H1 2026	Operating	36,000 / 675
Largest plant built (Stratos, Texas)	Start-up pending (end-2026 guided)	500,000
Largest announced hub (Project Cypress, Louisiana)	Announced, delayed	1,000,000
IEA Net Zero pathway, 2030	Modelled milestone	> 85,000,000
IEA Net Zero pathway, 2050	Modelled milestone	~ 980,000,000
Global fossil CO2 emissions, 2025	Context	38,100,000,000

Q3. Costs and the credibility of cost-reduction claims

F4. What buyers actually pay today is an order of magnitude above the cost-reduction endgame that motivates public funding. Climeworks charges individuals ~$1,000/t [12] ; analysis of the Orca plant puts its cost at $1,000-1,300/t [4] ; and CDR-market data show buyers becoming scarce above ~$500/t, with 2.47 Mt of DAC credits contracted by mid-2025 but only 1,186 t delivered [5] . The IEA estimates $125-335/t for a large-scale plant built today [1] . For comparison, the EU emissions-trading price is ~€65 (~$73) per avoided tonne [12] , and the main US subsidy pays $180/t for DAC with dedicated geologic storage [17] .

Projected costs cluster in a wide but far-from-$100 band. The foundational engineering estimate for a 1-Mt liquid-solvent plant gave $94-232/t (a design estimate, not a demonstrated plant) [2] ; the wider literature spans $100-1,000/t [3] . Independent ETH Zurich modelling (2024), built from component costs and expert assessment rather than simple learning curves, projects $230-540/t by 2050 for solvent systems, $280-580 for Climeworks-type solid filters and $230-835 for calcium-oxide designs, roughly twice as high as previously hoped, with no clear winner among technologies [4] . Even projected forward to a gigatonne of capacity, the cheapest design is expected to cost $226-544/t [5] . The credibility problem is structural: no commercial DAC plant has yet operated at its design capacity for long enough to demonstrate a learning curve (see F3), and the IEA's conditional view that costs could fall below $100/t by 2030 in the best locations [1] has no supporting operating record so far. A $100/t goal (the US "Carbon Negative Shot" target) should be treated as an aspiration, not a plan.

Table 2. Cost evidence for DAC, distinguished by basis.

Basis	Value (US$/t CO2)	Source character
Actually paid today	~500-1,300	Market data; plant analysis [4] [5] [12]
Large plant built today (estimate)	125-335	IEA assessment [1]

Basis	Value (US$/t CO2)	Source character
At-scale engineering estimate	94-232	Design estimate, undemonstrated [2]
Projected 2050 (independent)	230-540 (solvent); 280-580 (sorbent); 230-835 (CaO)	Expert-modelled projection [4]
US subsidy benchmark (45Q, storage)	180	Statutory [17]
Avoided tonne, EU ETS (for comparison)	~73	Market price [12]

Q4. Dependencies and integrity risks

Storage is a rate constraint, not a capacity constraint. Global geological capacity is ample for cumulative needs, but what matters is how fast CO2 can be injected with wells, permits and infrastructure: a 2024 Imperial College study puts the feasible benchmark at 5-6 Gt/yr globally by 2050 (an absolute maximum of ~16 Gt/yr only under an implausible US-dominated build-out), against 1.5°C-scenario removal requirements of 1-30 Gt/yr by mid-century across all CDR methods [15] . DAC's share is small within that, so storage is unlikely to be DAC's binding constraint, but storage permitting is a real schedule risk: Stratos only received its EPA Class VI injection permits in April 2025 [8] . Durability itself is well demonstrated where geology suits it: CO2 injected into Icelandic basalt was >95% mineralized in under two years in a monitored pilot [5] [12] .

Clean energy supply is the harder dependency. At 1,500-3,000 kWh/t, one gigatonne per year needs ~1,500-3,000 TWh/yr, about 5-10% of current world electricity demand (30,856 TWh in 2024); this is my arithmetic from the cited figures [5] [14] . That energy must be additional and low-carbon; otherwise the net-removal claim erodes proportionally, and gas-fired designs depend on recapturing their own combustion CO2 [3] [2] .

F5. Net-removal accounting and verification have already caught real shortfalls: that is the system working, but it is also a warning. Registry-based accounting forced the State of CDR assessment to revise its DAC delivery estimates downward once actual delivery data arrived [5] ; the same registries exposed the Orca and Mammoth shortfalls described in F3. Two accounting integrity risks deserve explicit attention: (i) under the July 2025 US tax law, CO2 used in enhanced oil recovery or products now earns the same $180/t credit as dedicated geologic storage [17] , blurring "removal" and "oil-field supply"; (ii) unaudited company performance claims have repeatedly diverged from verified delivery.

Demand and policy are fragile and highly concentrated. Microsoft represented ~90% of the carbon-removal market in 2025 and told suppliers in April 2026 that it was pausing new purchases (existing contracts continue): the industry's largest buyer stepping back, citing market conditions [11] . US federal support has whiplashed: mass award cancellations in October 2025, a proposed $2.3bn rescission of the hubs programme, set against retention of $1.2bn for the two flagship hubs and preservation of the $180/t credit [9] [10] [17] . Venture investment fell ~46% year-on-year in early 2025, and market watchers describe the industry as fragile, though Frontier's buyers committed a further $915m in June 2026 (total $1.8bn) [12] [18] . For any project that depends on selling credits, these are first-order risks, not background noise.

Q5. Implications for a pilot-funding decision

The arithmetic of pilot funding should be stated plainly. At today's verified prices (~$500-1,300/t actually paid), a €10m budget buys on the order of 10-20 kt of DAC removals (a hypothetical illustration, not a quotation), or 0.01-0.02% of a single gigatonne. A committee cannot buy climate-relevant volume; it can only buy learning: real cost, energy, utilization and reliability data from operating plants, which is precisely the evidence now missing (F3, F4). That reframing sets the conditions under which pilot funding makes sense:

1. Pay for delivered, independently verified tonnes: registry-verified removals, never nameplate capacity, with staged payments against milestones.
2. Require durable storage: dedicated geological storage or mineralization, not utilization or enhanced oil recovery, given the credit-parity risk in F5 [17] .
3. Require additional low-carbon energy and public disclosure of energy per tonne and utilization, so net-removal claims can be checked (F1) [3] .
4. Treat it as an R&D option, not procurement: time-limited, milestone-based with public performance reporting so each pilot teaches the sector [4] [5] .
5. Do not rely on credit sales as the exit : demand is concentrated and volatile (F5) [11] [12] .
6. Balance the portfolio : pair any DAC pilot with cheaper near-term removals.

The comparison with alternatives is brief but consequential. Conventional CDR (forestry, soil carbon, biomass burial) costs far less per tonne and already removes ~2.2 Gt/yr, but with lower durability and reversal risk; novel methods including biochar, BECCS and enhanced rock weathering span the middle; DAC is the most expensive but offers the highest measurement confidence and small land footprint [6] [1] . Cost estimates are not strictly comparable across methods (system boundaries differ), so the table is directional, not a procurement price list.

Table 3. Removal options compared (directional, from the 2026 State of CDR assessment as reported).

Option	Indicative cost (US$/t)	Durability
Forestry / restoration	5-53	Decades, reversal risk
Soil carbon	up to 150 (can be net-negative)	Years-decades, reversal risk
Biochar, BECCS, enhanced weathering	mid-range of novel methods	Decades-centuries
DAC with geological storage	~500-1,300 today; 226-580 projected	Millennia (mineralization/geologic)

The decision logic follows. If the committee's objective is maximum verified removals per euro now , the evidence favors conventional and mid-cost novel methods with strong verification, with DAC playing at most a token role. If the objective is building durable-removal capacity for mid-century , a conditional, verification-first DAC pilot is defensible as a learning investment under the six conditions above. The two objectives are compatible within one portfolio. In neither case should DAC tonnage be counted toward near-term climate targets: at any plausible pilot scale it is a rounding error, and every assessment reviewed treats CDR as a limited resource to be used alongside, never instead of, deep emissions cuts [6] [1] .

Limitations

This is a fast-moving field: several pipeline and demand figures may have changed after 9 November 2026. Company performance claims (Mammoth output, cost reductions) are unaudited and are labeled as such. Some primary documents were inaccessible (account-walled or blocked sites, including the full IEA report PDF and the State of CDR PDF beyond its front matter), so inspected reputable secondary accounts were used and identified. The IEA's facility counts date from 2022, and the deployment picture has shifted since, in both directions. Cost figures differ in system boundaries (capture vs capture-plus-storage vs net removal), so they are presented as ranges with bases rather than single points. This report's PDF was checked structurally and textually (page count, extracted text, links, bookmarks); no visual page inspection was available in the producing environment, so typography-related defects cannot be excluded. No specific region was specified, so evidence was weighed globally.

Conclusion

Q0, answered: DAC can plausibly become one component of mid-century carbon removal at the ~1 Gt/yr scale only under an unprecedented, sustained build-out: roughly a thousand 1-Mt-class plants by 2050, ~5-10% of today's world electricity in additional clean energy, and durable demand far beyond today's concentrated, volatile buyers. Today it operates six orders of magnitude below that benchmark, its flagship plants run far below nameplate, and the largest plant yet built has not started commercial operation on schedule. The answer is therefore possible in principle and in modelled pathways, but not demonstrated, not on track, and unlikely to be procured at climate-relevant scale on current evidence. Costs paid are ~$500-1,300/t; independent projections of $230-540/t by 2050 leave even the 2050 economics well above conventional alternatives.

For the committee: fund DAC pilots only as conditional learning investments (payment on verified delivered tonnes, durable storage, additional clean energy, public performance data, capped exposure), and expect the greater verified tonnage per euro from cheaper conventional and mid-cost removals in the near term. Deep emissions cuts remain the primary tool; DAC is a bet on the residual, hard-to-abate remainder of a net-zero world.

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